🤌 XAI Lecture 14

Network Dissection & TCAV

FAMAF - UNC

First Semester 2026

🧉 Disclaimer ---

This course is based on

Explainable Artificial Intelligence

From Simple Predictors to Complex Generative Models

Spring 2023, Harvard University

https://interpretable-ml-class.github.io/

🧉 Paper 1 ---

Network Dissection:
Quantifying Interpretability of Deep Visual Representations

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Abstract

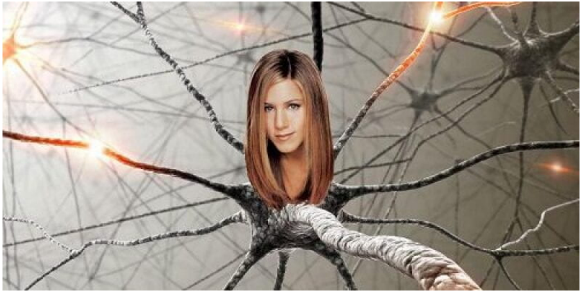
We propose a general framework called Network Dissection for quantifying the interpretability of latent representations of CNNs by evaluating the alignment between individual hidden units and a set of semantic concepts. Given any CNN model, the proposed method draws on a broad data set of visual concepts to score the semantics of hidden units at each intermediate convolutional layer. The units with semantics are given labels across a range of objects, parts, scenes, textures, materials, and colors. We use the proposed method to show that individual units are significantly more interpretable than random linear combinations of units, then

Figure 1. Unit 13 in [40] (classifying places) detects table lamps. Unit 246 in [11] (classifying objects) detects bicycle wheels. A unit in [32] (self-supervised for generating videos) detects people.

ance of a disentangled representation is not well-understood. A disentangled representation aligns its variables with a meaningful factorization of the underlying problem structure, and encouraging disentangled representations is a significant area of research [5]. If the internal representation of a deep

(Bau et al. 2017)

🧉 How Is Semantic Visual Concept Represented in the Brain? ---



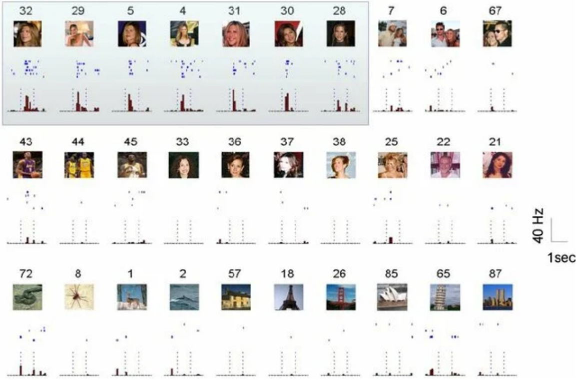
Neuroscientists have found that individual neurons can respond selectively to specific concepts — a property called **selectivity**.

The debate

Does the brain use **local** representations (one neuron = one concept) or **distributed** ones (concepts spread across many neurons)?

Question: Do deep neural networks learn similar structure?

A Neuron That Only Fires for Jennifer Aniston ---



Quiroga et al. (2005) recorded neurons in the human medial temporal lobe and found neurons that fire exclusively for specific people or landmarks.

- One neuron fired for Jennifer Aniston — photos, drawings, even her name written in text
- Another neuron fired only for the Eiffel Tower

Why this matters

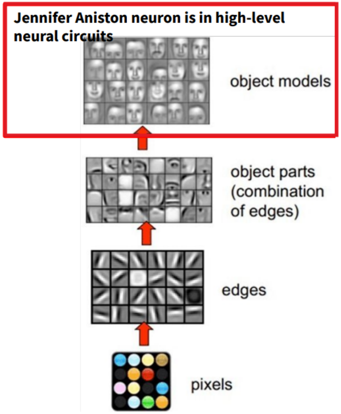
This is a **disentangled** representation: one neuron = one concept. The paper asks whether CNNs learn something analogous.

🧉 Disentangled Representation in Visual Cortex ---

The visual cortex processes information in a **hierarchy** — the **ventral stream** (“what pathway”):

- **V1**: edges, orientations (small receptive fields)
- **V2/V4**: textures, shapes, color patterns
- **IT** (inferotemporal): objects, faces — where the “Jennifer Aniston neuron” lives

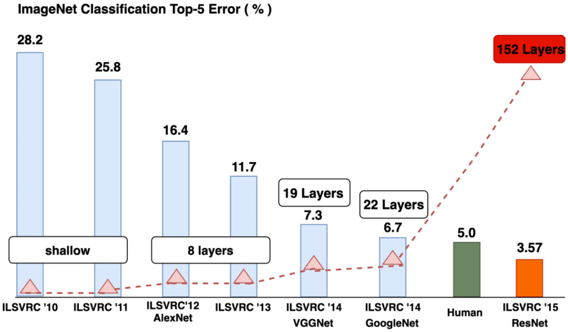
Along this hierarchy: **selectivity** increases and **invariance** increases.



Bridge to CNNs

This layered specialization is the biological inspiration for deep CNNs. Network Dissection tests whether CNNs develop analogous detectors at each layer.

🧉 Deep CNN for Computer Vision ---



CNNs revolutionized computer vision, reaching superhuman performance on ImageNet — but they remain largely opaque.

- Units in early layers respond to edges and colors
- Units in later layers seem to respond to objects and scenes (Zhou et al., 2015)
- But this was observed qualitatively — no rigorous measurement existed

We know CNNs *work* extremely well. We don't know *how*, or what each unit has learned. Network Dissection provides the first quantitative answer.

🧉 Proposed Questions ---

The paper frames the problem around three concrete research questions. Each has a direct answer by the end of the paper.

1. **What is a disentangled representation, and how can its factors be quantified and detected (in deep CNN)?**
2. **Do interpretable hidden units reflect a special alignment of feature space, or are interpretations a chimera?**
3. **What conditions in state-of-the-art training lead to representations with greater or lesser entanglement?**

🧉 Proposed Questions and Contributions ---

Q1: How to quantify disentangled representations?

- Proposed a metric, intersection over union score (IoU), to quantify the interpretability of each unit
- The alignment level between unit activated area and human-interpretable concepts

Q2: Are interpretable units real or a chimera?

- A semantic concept can be detected by many units
- A unit can detect many semantic concepts
- But the natural basis is **special**: rotating it destroys interpretability

Q3: What training conditions affect entanglement?

- Factors tested: layer depth, training iterations, the angle of the images, input datasets, dropout, batch normalization, supervised vs. self-supervised, layer width

🧉 Related Works ---

Prior work tried to understand CNN internals through **visualization**, but all approaches share a fundamental limitation: they are qualitative and cannot be used to rigorously compare models.

- **Generative Visualizations**: Synthesize images that maximally activate a unit (Mahendran et al., 2015; Nguyen et al., 2016; Simonyan et al., 2014)
- **Salience-based Visualizations**: Highlight which pixels most contribute to a unit's activation (Zeiler et al., 2014)
- **Global Analysis**: Project the full representation space into 2D for inspection (t-SNE: van der Maaten et al., 2008; Yosinski et al., 2015)

Common limitation

All these methods produce images that a **human** must then interpret — subjective and impossible to compare across networks. Network Dissection replaces the human with a **metric**.

🧉 Broden: Broadly and Densely Labeled Dataset ---

To measure interpretability we need a **ground truth** of human concepts. Broden unifies five existing segmentation datasets into a single resource.

Figure 2. Samples from the **Broden** Dataset. The ground truth for each concept is a pixel-wise dense annotation.

Category	# Classes
Scene	468
Object	584
Part	234
Material	32
Texture	47
Color	11

Multiple labels can apply to the **same pixel** (e.g., a black cat leg → “cat”, “leg”, “black”)

The range from colors to scenes allows testing nterpretability at **every level of abstraction**.

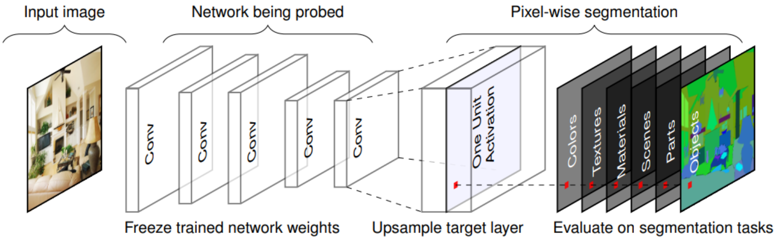
🧉 Scoring Unit Interpretability ---

Pipeline:

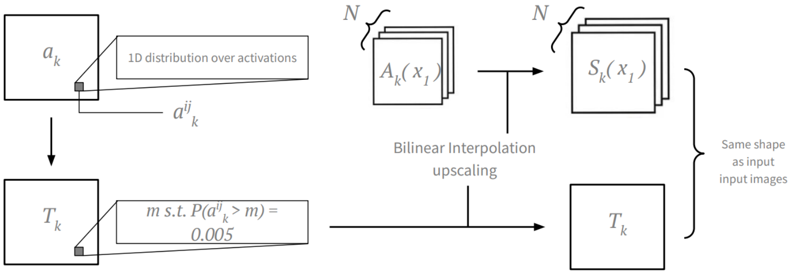
- Feed images from Broden through the frozen CNN
- Collect activation map $A_k(x)$ for each unit k
- Upsample to input resolution $\rightarrow S_k(x)$
- Threshold to obtain a binary mask $M_k(x)$
- Compare against concept labels L_c

Each convolutional unit k acts as a **spatial filter**, producing an activation map over the image. The key idea is to treat each unit as a candidate **segmentation model** for one concept.

Each unit is evaluated against all **1,197 Broden segmentation tasks**, and is assigned the concept with the highest score.



🧉 Scoring Unit Interpretability: Threshold ---



Threshold T_k: value such that $P(a^k_{ij} > T_k) = 0.005$

This means: only the **top 0.5%** of activations for unit k across the entire dataset are considered "active."

Why a quantile?

Each unit has a different activation range. A fixed threshold would favor high-magnitude units. The quantile normalizes this.

🧉 What Does "Accuracy" Mean Here? ---

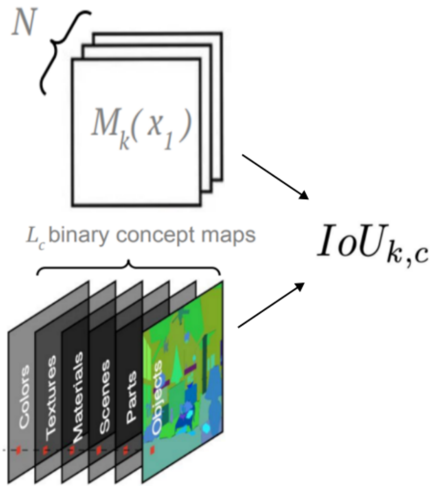
Each unit is treated as a **pixel-wise detector**.

- The task is **not** image classification
- The task is **not** concept presence
- The task is:
  "When this unit activates, does it activate on pixels that humans label as concept c?"

Key idea

IoU measures **spatial alignment**, not prediction performance.

🧉 Scoring Unit Interpretability: IoU ---



$$IoU_{k,c} = \frac{\sum_x |M_k(x) \cap L_c(x)|}{\sum_x |M_k(x) \cup L_c(x)|}$$

In plain English

"Of all the pixels where **either** the unit fires **or** the concept is present, what fraction has **both**?"

Detector: Unit k is a detector for concept c if $IoU_{k,c} > 0.04$

🧉 Quantifying Alignment: Summary ---

The interpretability of unit k for concept c is the **IoU** across the dataset:

- **Detector**: Unit k is a detector for concept c if $IoU_{k,c} > 0.04$
- **Top Label**: If a unit matches multiple concepts, the one with the highest IoU is assigned
- **Layer Score**: The interpretability of a layer is the number of **unique** semantic concepts aligned with its units

Why "unique"?

If 50 units all detect "dog", the layer score counts "dog" only once. This measures **diversity** of concepts, not just quantity of detectors.

This is not a performance metric.

🧉 Experiments: Tested CNN Models ---

The paper evaluates interpretability across diverse architectures and training regimes to identify which conditions favor disentangled representations.

Training	Network	Dataset / Task
None	AlexNet	Random weights
Supervised	AlexNet, GoogLeNet, VGG-16, ResNet-152	ImageNet, Places205, Places365, Hybrid
Self-supervised	AlexNet	10 proxy tasks (context, puzzle, colorization, etc.)

Places: scene-centric dataset with categories such as kitchen, living room, coast.

Scene-centric vs. Object-centric

Models trained on **Places** (scenes) enerally develop more object detectors than those trained on **ImageNet** (objects). Why? Because recognizing a scene *requires* identifying the objects in it.

🧉 Experiment 1: Human Evaluation of Interpretations ---

- Does Network Dissection agree with **human judgment**?
  - Raters shown 15 images with highlighted patches for each unit in AlexNet (Places205)
  - Asked: "Does this phrase describe most of the patches?" (yes/no)
  - **Network Dissection accuracy**: portion of method's labels rated as descriptive
  - **Human consistency**: portion of ground-truth labels found descriptive by a second group of raters

	conv1	conv2	conv3	conv4	conv5
Interpretable units	57/96	126/256	247/384	258/384	194/256
Human consistency	82%	76%	83%	82%	91%
Network Dissection	37%	56%	54%	59%	71%

Two takeaways

1. Network Dissection agreement **increases** in higher layers (37% $\rightarrow$ 71%)
2. Human consistency remains high throughout (76–91%), setting an upper bound

🧉 Experiment 2: Axis-Aligned Interpretability ---

Is interpretability a property of the **natural basis** learned by the network, or does it appear in **every direction**?

Hypothesis 1: Concepts appear in every direction

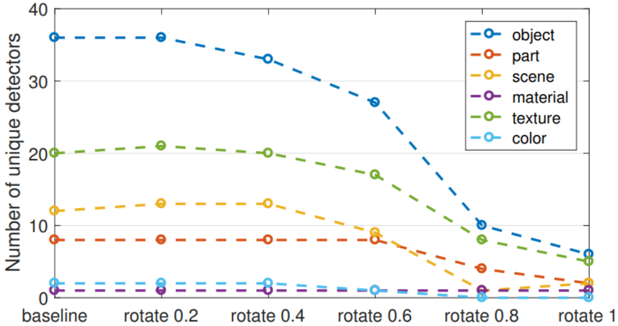
- Single units are not more interpretable than random combinations
- Interpretability would be an artifact, not a real property

Hypothesis 2: The natural basis is special

- The model converges to a semantically rich, axis-aligned basis
- Interpretability is a learned property of the representation

Test: Apply a random orthogonal rotation R to the representation space. If H2 is correct, rotating the basis should **destroy** interpretability.

🧉 Experiment 2: Rotation Results ---

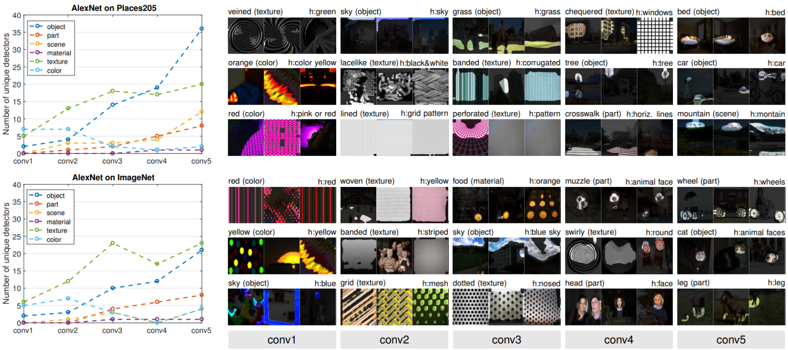


Number of unique detectors **decreases up to 80%** as the basis is rotated away from the natural one.

Conclusion

Interpretability is **not** an inevitable byproduct of discriminative power. It is a **special alignment** learned by the network — **Hypothesis 2 confirmed**.

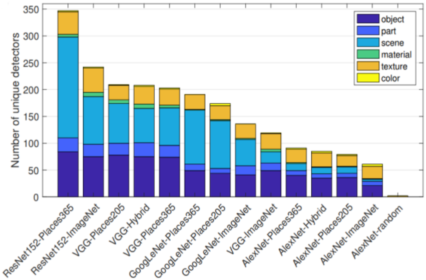
🧉 Experiment 3: Concepts by Layer ---

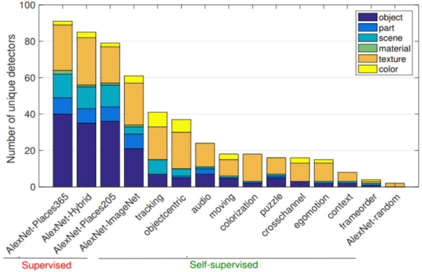


The type of concept detected changes with **layer depth**: **Early layers**: colors, textures → **Middle layers**: materials, parts → **Late layers**: objects, scenes.

This mirrors the visual cortex hierarchy: V1 (edges) → V4 (textures) → IT (objects). The analogy holds quantitatively.

Experiment 4: Network Architectures ---





Comparing the number of unique detectors across architectures trained on the **same dataset**:

- Deeper and more complex architectures tend to develop **more unique concept detectors**
- **ResNet-152** and **VGG-16** consistently outperform AlexNet and GoogLeNet

Key nuance

More detectors does not always mean better accuracy — the relationship is architecture-dependent.

🧉 Experiment 5: Training Conditions ---

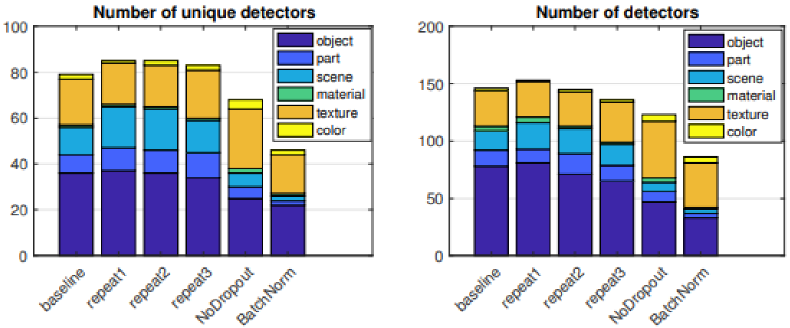
What training choices affect interpretability?

- **Weight initializations**: **Minimal effect** — models converge to similar levels of interpretability regardless of random seed
- **Dropout**: **Some effect** — removing dropout leads to more "texture" detectors and fewer "object" detectors
- **Batch Normalization**: **Significant effect** — interpretability decreased substantially

Surprising result

Batch Normalization improves accuracy but **hurts** interpretability. There may be a trade-off between optimization convenience and semantic disentanglement.

🧉 Experiment 5: Training Conditions Results ---



The figure confirms:

- Different random seeds → nearly identical interpretability
- Dropout removal → shift from objects to textures
- Batch Norm → large drop in unique detectors across all categories

🧉 Experiment 6: Discrimination vs. Interpretability ---

Does being more "interpretable" make a model **better at new tasks**?

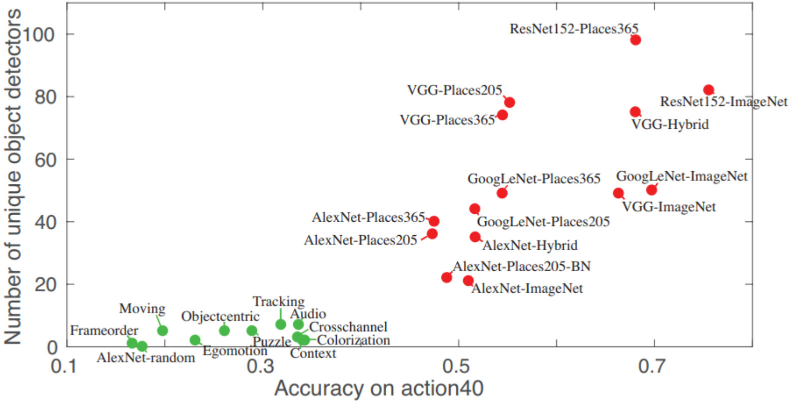
Benchmark:

- Across several CNNs, extract activations from high layers
- Train a **linear SVM** on a new task: **action recognition** (Action40 dataset)
- Compare classification accuracy vs. number of unique object detectors

Why a linear SVM?

A linear classifier can only succeed if the features are already well-organized. If interpretable representations also produce linearly separable features, that's strong evidence they generalize.

🧉 Experiment 6: Discrimination Results ---



Result: Positive correlation between object detectors and classification accuracy.

Implication Encouraging concept detection can improve discrimination. Interpretability and performance are not at odds, they can reinforce each other.

🧉 Experiment 7: Layer Width ---

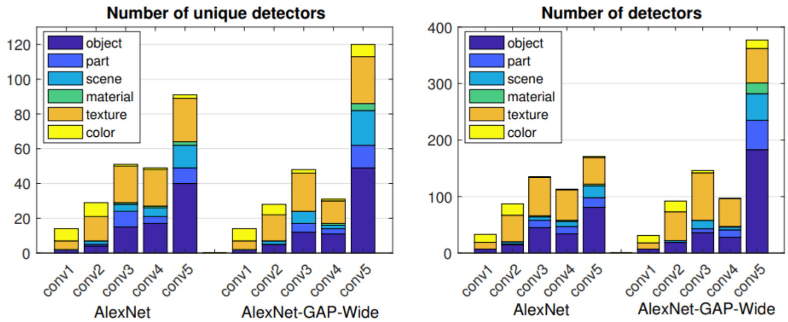
What happens if we make a layer **wider** (more units)?

- **Method**: Tripled the number of units in AlexNet's conv5 (from 256 to 768)
- **Result**: Accuracy remains similar, but the number of **unique concept detectors** increases significantly
- **Limit**: Beyond ~1024 units, diminishing returns in unique concepts

Intuition

Wider layers provide more "slots" for the network to separate explanatory factors — like having a bigger bookshelf with room for more categories.

🧉 Experiment 7: Width Results ---



- Increased width:
  - More unique detectors both at the widened layer **and** in the network generally
  - Accuracy remains stable
  - Effect saturates beyond a threshold
- If interpretability is a goal, **wider layers** are a cheap way to get more disentangled representations without sacrificing accuracy.

🧉 Discussion Questions (Paper 1) ---

1. **Distribution Understanding:** Concept detectors from a particular dataset betray something about the underlying distribution. How do you think this can be applied in the real world (e.g., bias detection)?
2. **Single unit to circuit:** This method interprets single units; could it be extended to larger circuits and would this be useful?
3. **Beyond vision:** This method is deeply tied to the vision domain. Could it be extended to other domains such as natural language?

🧉 Paper 2 ---

Interpretability Beyond Feature Attribution: Quantitative Testing with Concept Activation Vectors (TCAV)

Been Kim Martin Wattenberg Justin Gilmer Carrie Cai James Wexler Fernanda Viegas Rory Sayres

Abstract

The interpretation of deep learning models is a challenge due to their size, complexity, and often opaque internal state. In addition, many systems, such as image classifiers, operate on low-level features rather than high-level concepts. To address these challenges, we introduce Concept Activation Vectors (CAVs), which provide an interpretation of a neural net's internal state in terms of human-friendly concepts. The key idea is to view the high-dimensional internal state of a neural net

A key difficulty, however, is that most ML models operate on features, such as pixel values, that do not correspond to high-level concepts that humans easily understand. Furthermore, a model's internal values (e.g., neural activations) can seem incomprehensible. We can express this difficulty mathematically, viewing the state of an ML model as a vector space E_m spanned by basis vectors e_m which correspond to data such as input features and neural activations. Humans work in a different vector space E_h spanned by implicit vectors e_h corresponding to an unknown set of human-interpretable concepts.

(Kim et al. 2018)

🧉 From Network Dissection to TCAV - ver si lo dejo ---

Network Dissection (Bau et al., 2017)

- Interprets **individual units**
- Concepts come from a **fixed dataset** (Broden)
- Measures: does unit k detect concept c?
- Spatial alignment (IoU)
- Limited to CNNs with spatial activations

TCAV (Kim et al., 2018)

- Interprets **entire classes**
- Concepts defined by the **user** (any set of examples)
- Measures: is concept c important for class k?
- Directional sensitivity in activation space
- Works on any differentiable model

The shift

Network Dissection asks: "what does this unit detect?"
TCAV asks: "how important is this concept for this prediction?"

🧉 Motivation + Problem Statement ---

- Interpreting deep learning models is crucial to understanding their behavior and ensuring accurate predictions
- But remains a big challenge due to size, complexity, and opacity of ML models
- Many systems operate on **low-level features** (e.g., pixel values) rather than **high-level concepts** (e.g., face) that are human-interpretable

The mismatch

Models explain in **pixels**. Humans understand in **concepts**. TCAV bridges this gap.

Motivation + Problem Statement ---

Problem:

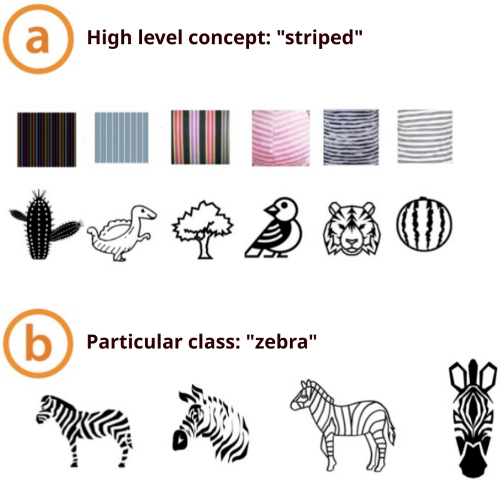
- We can't express these concepts as pixels
- And they weren't our input features
- Existing methods (saliency maps, LIME) explain in terms of input features — not the concepts humans care about

What we want to ask

"Did the model use the concept of *person*? Of *money*? Of *machine*?" — not "which pixels mattered?"

🧉 Summary of Contributions ---

- Introduce **Concept Activation Vectors (CAVs)**: way to interpret a neural network's internal state in terms of human-friendly concepts
- Key idea: use the high-dimensional internal state of a neural net as an **aid**, not an obstacle
- Main contribution: **Testing with CAV (TCAV)**, that quantifies model sensitivity to a high-level concept learned by a CAV for a particular class



Example

"How sensitive is the class *zebra* to the concept *striped*?" $\rightarrow$ TCAV returns a single number.

🧉 Goals of TCAV ---

- **Accessibility:** requires little to no ML expertise
- **Customization:** adaptable to any concept, even outside of training
- **Plug-in readiness:** works without retraining/modifying ML models
- **Global quantification:** can interpret entire classes with a single quantitative measure

↓

Assessed with experiments + human evaluation

Compare with Network Dissection

ND is not accessible (requires understanding IoU), not customizable (fixed Broden concepts), and not global (per-unit, not per-class). TCAV addresses all three.

🧉 Related Work: Limitations of Current Methods ---

Perturbation-based (LIME, SHAP):

- Explain individual predictions (**local**), not entire classes
- Operate at the input feature level, not at the concept level

Saliency methods (Gradient, CAM, etc.):

- **Local**: one image at a time
- **Lack customization**: cannot test user-defined concepts
- **Vulnerable** to adversarial attacks (Ghorbani et al., 2017)
- **Insensitive** to model randomization (Adebayo et al., 2018)

Core limitation shared by all

These methods explain in terms of **input features**. Saliency maps tell you *where* the model looks, not *what* it sees. TCAV explains in terms of **human-defined concepts**.

🧉 Related Work: Linearity + Latent Dimensions ---

A key observation across deep learning: **concepts tend to be linear directions** in representation spaces.

- **word2vec** (Mikolov et al., 2013): $\vec{king} - \vec{man} + \vec{woman} \approx \vec{queen} \rightarrow$ “gender” is a linear direction
- **Probing classifiers** (Alain & Bengio, 2016): linear classifiers on hidden layers extract semantic information
- **Network Dissection** (Bau et al., 2017): individual units align with concepts — the axes themselves are meaningful

TCAV builds on this

If concepts are **linear directions** in activation space, we can find them with a simple linear classifier and measure their influence via **directional derivatives**.

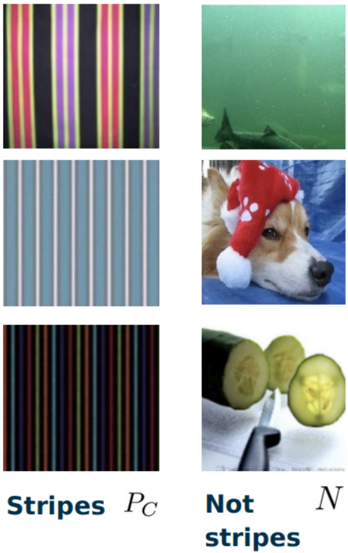
🧉 Approach: Defining the CAV ---

The user wants to test concept C (e.g., "stripes").
How do we represent it inside the network?

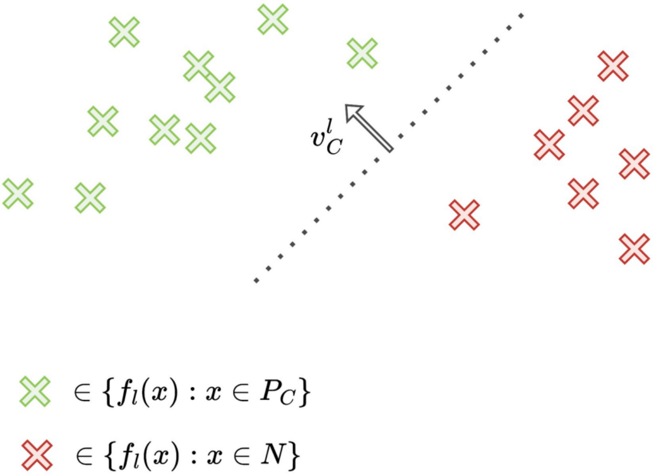
1. Collect a set of examples P_C of the that concept and a negative set N of examples that don't
2. Feed both through the network, extract activations at layer l
3. Train a **linear classifier** to separate them
4. The **CAV** $\mathbf{v}_C^l$ is the vector **orthogonal** to the decision boundary between activations — it points in the "direction of the concept"

Key insight

A CAV is a **direction** in activation space that represents a human-defined concept. No retraining of the original model needed.



🧉 Approach: Visualizing the CAV ---



The CAV $\mathbf{v}_C^l$ is the normal to the linear decision boundary:

- One side $\to$ "has concept C"
- Other side $\to$ "does not have concept C"

Moving along the CAV direction = "adding more of concept C" to the representation.

🧉 Approach: Concept Sensitivity ---

Saliency maps gauge sensitivity with respect to per-pixel perturbations. With CAV, we gauge sensitivity **towards a concept** at an entire layer:

$$S_{C,k,l}(\mathbf{x}) = \nabla h_{l,k}(f_l(\mathbf{x})) \cdot \mathbf{v}_C^l$$

In plain English

"If we nudge the activations in the direction of concept C, how much does the prediction for class k change?"

- $f_l(\mathbf{x})$: activations at layer l for input $\mathbf{x}$
- $\nabla h_{l,k}(f_l(\mathbf{x}))$: gradient of class k output w.r.t. those activations
- $\mathbf{v}_C^l$: the CAV (concept direction)
- The **dot product** measures alignment between the gradient and the concept direction

🧉 Approach: The TCAV Score ---

The sensitivity $S_{C,k,l}(\mathbf{x})$ is **per-image**. To get a **global** measure for the entire class:

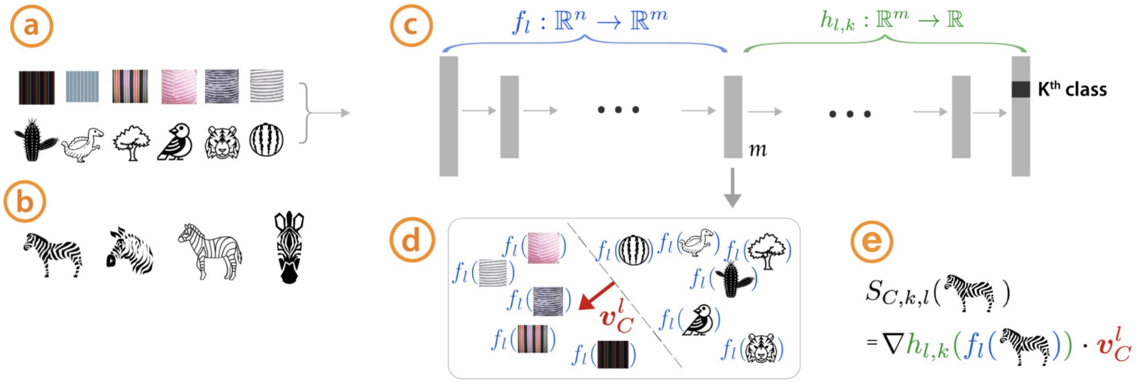
$$\text{TCAV}_{Q_{C,k,l}} = \frac{|\{\mathbf{x} \in X_k : S_{C,k,l}(\mathbf{x}) > 0\}|}{|X_k|}$$

In plain English

"Of all images of class k, what fraction has activations that are **positively influenced** by concept C?"
- TCAV $= 1.0 \rightarrow$ every image of class k is sensitive to C
- TCAV $= 0.5 \rightarrow$ no more influence than random (meaningless)
- TCAV $= 0.0 \rightarrow$ concept C pushes **away** from class k

Statistical safeguard: A t-test across multiple random negative sets filters spurious CAVs. Only statistically significant results are reported.

Approach Summary ---



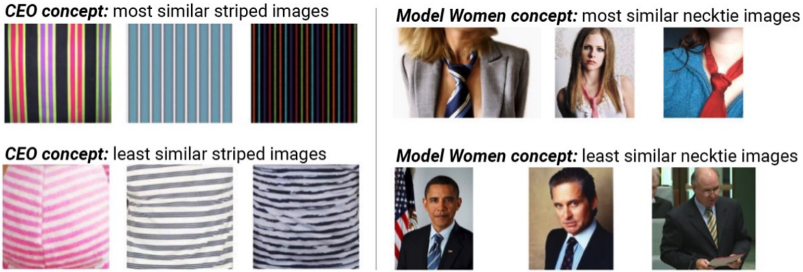
The full pipeline in one sentence: User provides concept examples $\to$ linear classifier on activations $\to$ CAV direction $\to$ directional derivative $\to$ TCAV score $\to$ t-test.

The original model is never modified.

🧉 Results: Sorting Images with CAVs ---

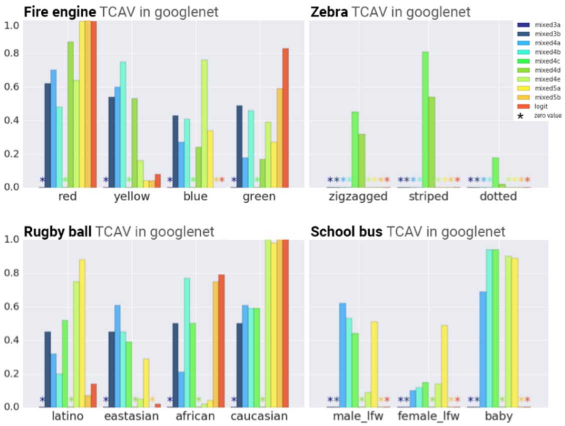
Images sorted by their projection onto a CAV direction, a qualitative validation that the CAV captures the intended concept.

- Class "stripes" sorted by concept "CEO": confirms CAV reflects the concept correctly
- Class "necktie" sorted by concept "model woman": reveals a gender bias in the learned representation



Sorting by unexpected concepts can reveal biases invisible to standard evaluation metrics.

🧉 Results: Gaining Insights with TCAV ---



Each bar = TCAV score at a different layer.
Only statistically significant CAVs shown.

- "Red" → "fire engine": high TCAV
- "Striped" → "zebra": high TCAV
- "Dotted" → "zebra": filtered out by t-test
- "Caucasian" → "rugby ball": high TCAV
-
  - TCAV enables **ranking concepts by importance** for a class — not just "is it relevant?" but "how much more relevant than others?"

Layers closer to the output have greater influence on the prediction.

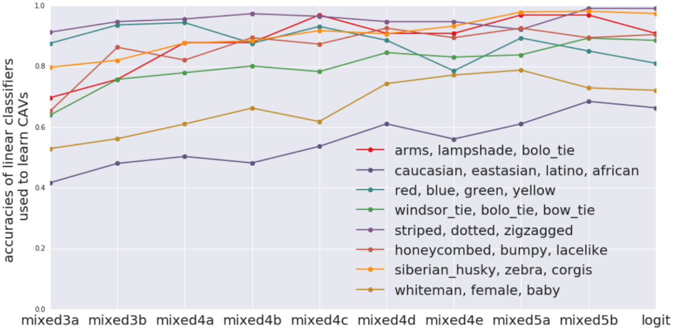
Key result

TCAV confirms intuitive associations **and** reveals hidden biases — with statistical rigor.

🧉 Results: TCAV for Where Concepts Are Learned ---

CAV accuracy at different layers reveals *where* each concept is learned:

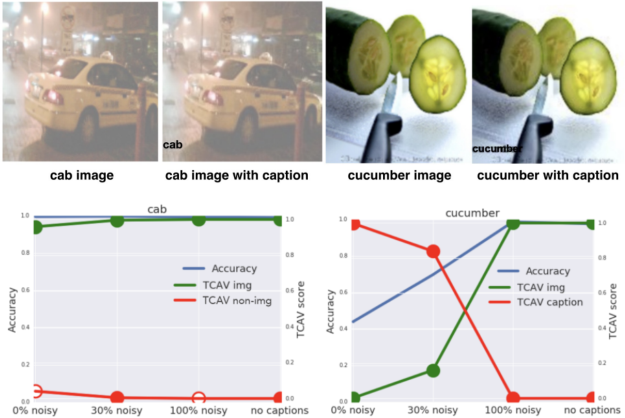
- Simple concepts (colors, patterns) reach high accuracy at low layers.
- Complex concepts (age, sex, objects) don't reach high accuracy until higher layers.



Convergence with Network Dissection

Both methods confirm the same hierarchy using completely different approaches: early layers = low-level features, late layers = high-level concepts.

🧉 Results: Controlled Experiment with Ground Truth ---



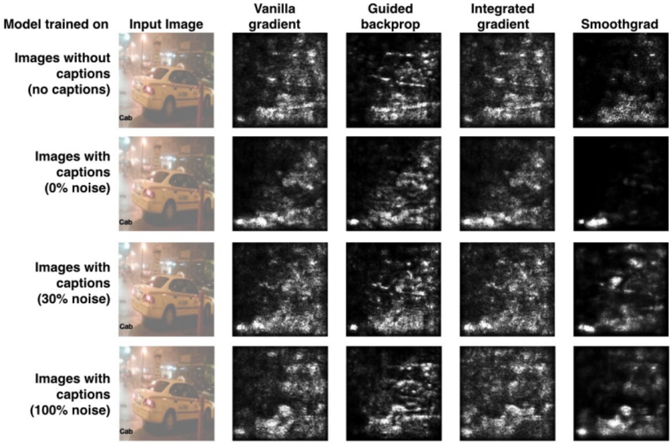
A model trained on images with **embedded captions** to create known ground truth:

- **"Cab"**: image concept dominates → TCAV correctly shows high image sensitivity, low caption sensitivity
- **"Cucumber"**: caption concept dominates when present → TCAV correctly reflects this

Validation

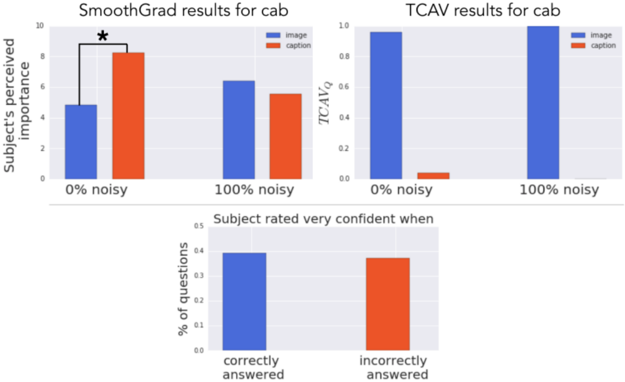
TCAV scores faithfully reflect the **true** importance of each concept — confirmed against known ground truth.

🧉 Results: Evaluation of Saliency Maps ---



We know that for "cab" the image concept is most important. But saliency maps highlight the caption text (high contrast) instead — across all four methods.

🧉 Results: Saliency Maps Are Misleading ---

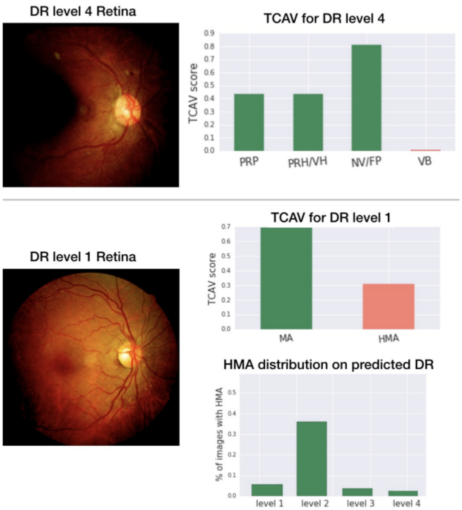


Using the same controlled experiment:

- **Saliency maps** led human subjects to **incorrect conclusions** about which concept was more important
- Subjects thought the caption was more important than the image for "cab" — **wrong**
- **TCAV scores** correctly identified the dominant concept in every case

Saliency maps are not just imprecise — they are actively **misleading**. They highlight visually salient pixels (high contrast text), not causally important ones.

🧉 Results: TCAV for Medical Diagnosis ---



TCAV applied to a diabetic retinopathy diagnosis model:

- Green: medically relevant concepts
- Red: irrelevant concepts
- **Level 4**: model relies on medically relevant concepts
- **Level 1**: model incorrectly relies on HMA (relevant for Level 2, not Level 1)

Debugging with TCAV

A doctor (no ML expertise) can verify whether the model uses the **right medical concepts**, not just whether it predicts the correct label.

🧉 Conclusions ---

Roadmap: Gradient → Attention → Concept-based (**TCAV!**)

Limitations:

- Evaluated only on computer vision tasks
- Assumes concepts are **linear** directions in activation space
- Statistical significance testing — is a t-test rigorous enough?
- Depends on quality of user-provided concept examples

🧉 Discussion Questions (Paper 2) ---

- How does this method compare to e.g., saliency maps?
- How would you adapt TCAV for other types of data (e.g., audio/video) and what would that look like?
- Thoughts on TCAV for adversarial example identification (Appendix A)?
- Do you think there could be adversarial images that allow meaningless CAVs to pass the statistical significance test?

Thank You!

References --- |

Bau, David, Bolei Zhou, Aditya Khosla, Aude Oliva, and Antonio Torralba. 2017. "Network Dissection: Quantifying Interpretability of Deep Visual Representations." In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition*, 6541–49.

Kim, Been, Martin Wattenberg, Justin Gilmer, Carrie Cai, James Wexler, Fernanda Viegas, and Rory Sayres. 2018. "Interpretability Beyond Feature Attribution: Quantitative Testing with Concept Activation Vectors (TCAV)." In *International Conference on Machine Learning*, 2668–77.